

Confidence envelopes for the false discoveries with heterogeneous data

Romain Périer, joint work with Gilles Blanchard and Guillermo Durand.

Multiple testing setting

Formal setting of multiple testing - notations

Let $X \in \mathcal{X}$ be a sample of distribution $\mu \in \mathcal{P}$ where $\mathcal{P}$ is a set of probability measures. Define :

- $H_1, \dots, H_m \subset \mathcal{P}$, m null hypotheses,
- For all i , a p -value $p_i(X)$ such that, if $\mu \in H_i$, $F_i(\alpha) := \mathbb{P}_{X \sim \mu}(p_i \leq \alpha) \leq \alpha$ for all $\alpha \in [0, 1]$,

Goal : find a "good" rejection set $R(X)$ under some error constraints.

Notations :

- $\mathcal{H}_0(\mu) = \{i : \mu \in H_i\}$,
- $m_0 = |\mathcal{H}_0(\mu)|$,
- $V(R) = |\mathcal{H}_0(\mu) \cap R|$.

Classic error constraints

Let R be a rejection set. Then we define the Family-Wise Error Rate (FWER) as

$$\text{FWER}(R) = \mathbb{P}(V(R) > 0).$$

Example 1 (Bonferroni procedure)

Let $\alpha \in [0, 1]$ and $R^{\text{Bonf}} = \{i \in \llbracket 1, m \rrbracket : p_i \leq \frac{\alpha}{m}\}$. Then $\text{FWER}(R^{\text{Bonf}}) \leq \alpha$.

We define the False Discovery Rate (FDR) as the mean of the False Discovery Proportion (FDP):

$$\text{FDP}(R) = \frac{V(R)}{|R| \vee 1}, \quad \text{FDR}(R) = \mathbb{E}[\text{FDP}(R)].$$

Example 2 (Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995))

Let $\alpha \in [0, 1]$, $\hat{k}^{\text{BH}} = \max\{k : p_{(k)} \leq \frac{\alpha k}{m}\}$ and $R^{\text{BH}} = \{i \in \llbracket 1, m \rrbracket : p_i \leq \frac{\alpha \hat{k}}{m}\}$. Then, if $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values and is independent of $(p_i)_{i \in \mathcal{H}_1}$, $\text{FDR}(R^{\text{BH}}) \leq \alpha$.

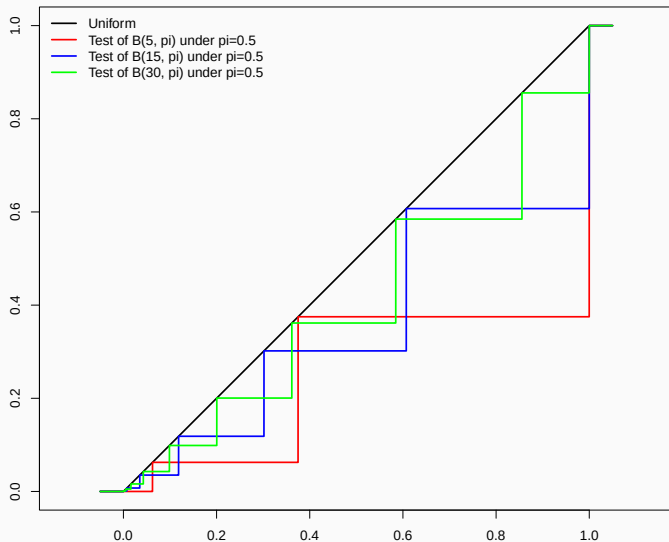
Binomial tests : Assume $X = (X_1, \dots, X_m)$, where $(X_i)_{1 \leq i \leq m}$ are independent variables such that $X_i \sim \mathcal{B}(n_i, \pi_i)$ under H_i . Then :

- $p_i(X_i) := \mathbb{P}_{Y \sim \mathcal{B}(n_i, \pi_i)}(Y \leq X_i) = \sum_{j=0}^{X_i} \binom{n_i}{j} \pi_i^j (1 - \pi_i)^{n_i-j}$ is a p -value for i ,
- For all i , p_i is a discrete p -value and if the couples (n_i, π_i) are different, the respective cdf F_i are.

Other examples:

- Fisher's exact tests,
- Genome-wide association study.

CDFs of discrete and uniform p -values



Confidence envelopes

Definition 1 (Confidence envelope)

Let $\alpha \in (0, 1)$. A confidence envelope of level α is a statistic $\hat{V}_\alpha : \mathcal{P}(\mathbb{N}_m) \rightarrow \mathbb{N}$ such that, for all $\mu \in \mathcal{P}$,

$$\mathbb{P}_{X \sim \mu}(\forall S \subset \mathbb{N}_m, V(S) \leq \hat{V}_\alpha(S)) \geq 1 - \alpha.$$

- Allows user to choose freely their rejection set.

Definition 2 (Joint Error Rate)

For a reference family $\mathfrak{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ and $\mu \in \mathcal{P}$, the joint error rate is :

$$JER(\mathfrak{R}, \mu) := \mathbb{P}_{X \sim \mu}(\exists k \in \mathbb{N}_K : V(R_k) > \zeta_k).$$

- Confidence envelope ideas restricted at our reference regions,
- Allows to compute a valid confidence envelope with interpolation (Blanchard et al., 2020, Durand et al., 2020).

In the following, we will be in a **top- k setting** :

$$R_k = \{\sigma(i), 1 \leq i \leq k\} \text{ (where } p_{\sigma(1)} \leq \dots \leq p_{\sigma(m)}).$$

Bretagnolle's inequality

Proposition 1 (DKWM inequality - Improved version from Reeve, 2024)

Let $X_1, \dots, X_m \stackrel{iid}{\sim} F$ and $\hat{F}_m(t) := \frac{1}{m} \sum_{1 \leq i \leq m} \mathbb{1}_{X_i \leq t}$ for all $t \in \mathbb{S}$.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} (\hat{F}_m - F)(t) > \lambda\right) \leq \exp(-2\lambda^2).$$

When $(p_i)_{1 \leq i \leq m} \stackrel{iid}{\sim} \mathcal{U}([0, 1])$, $F(t) = t$.

Proposition 2 (Bretagnolle's inequality, Shorack and Wellner, 1986 with the improvement from Reeve, 2024)

Let $X_1, \dots, X_m$ be independent, denote by F_i the cdf of X_i and $\bar{F}$ the mean of F_i 's.

Then, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{m} \sup_{t \in \mathbb{R}} (\hat{F}_m - \bar{F})(t) > \lambda\right) \leq e \cdot \exp(-2\lambda^2).$$

Theorem 1 (Bretagnolle's top- k calibration)

Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values. Define

$$\zeta_k = k \wedge \left\lfloor m\bar{F}(p_{(k)}) + \sqrt{m}\lambda_\alpha \right\rfloor.$$

Then, the reference family $\mathfrak{R} = ((R_k, \zeta_k))_{k \in \mathcal{K}}$ has its JER bounded by α .

Theorem 2 (Adaptive version of Bretagnolle's top- k calibration)

Let $\alpha \in (0, 1]$ and $\mathbb{G}(t) = (\mathbb{1}_{\{p_i \leq t\}} - F_i(t))_{1 \leq i \leq m}$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values. Define $\lambda_\alpha = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$, $\hat{m}_0$ such that

$$\left\{ \sup_{t \in [0, 1]} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \sqrt{m_0} \lambda_\alpha \right\} \subset \{m_0 \leq \hat{m}_0\}, \text{ and}$$

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \leq i \leq \hat{m}_0} \left(\mathbf{F}(p_{(k)}) \right)_{[i]} + \sqrt{\hat{m}_0} \lambda_\alpha \right\rfloor^1.$$

Then, the reference family $\mathfrak{R} = ((R_k, \zeta_k))_{k \in \mathcal{K}}$ has its JER bounded by α .

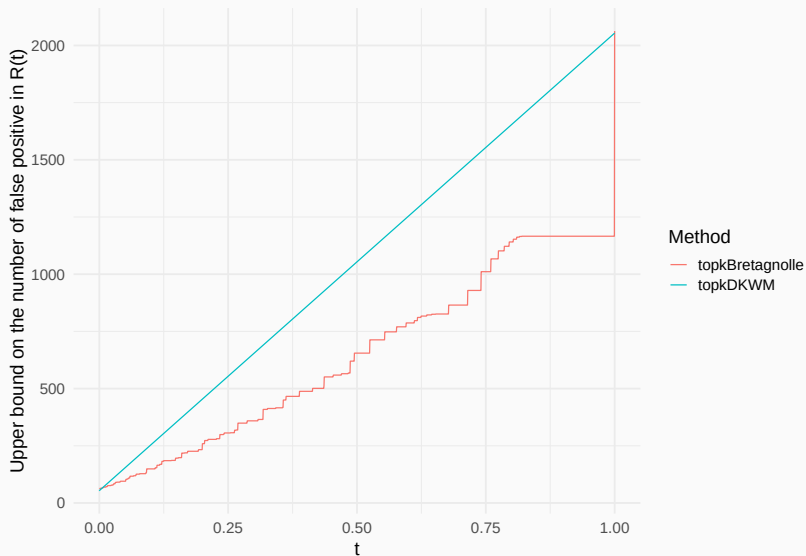
Sketch of proof:

- $|R_k \cap \mathcal{H}_0| \leq \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq p_{(k)}}$,
- Evaluate Bretagnolle's inequality at point $p_{(k)}$,
- Bound $\sqrt{m_0}$ by $\sqrt{\hat{m}_0}$ and $\sum_{i \in \mathcal{H}_0} F_i(p_{(k)})$ by $\sum_{1 \leq i \leq \hat{m}_0} \left(\mathbb{F}(p_{(k)}) \right)_{[i]}$ (worst case).

¹ $x_{[i]}$ is the i -th largest coordinate of x .

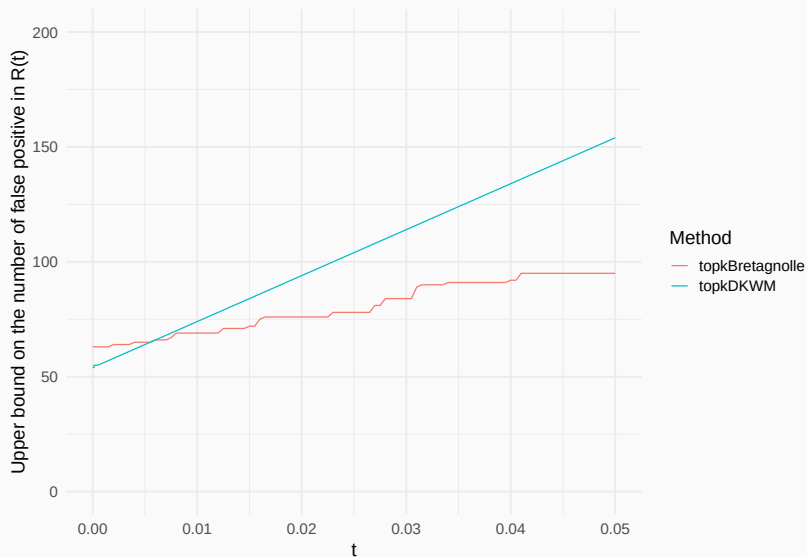
Simulations (1/3)

Comparison of bounds efficiency

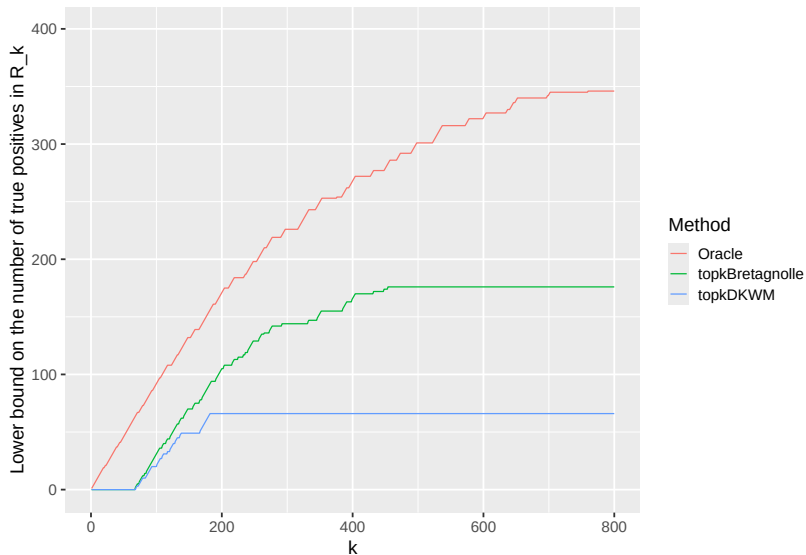


Simulations (2/3)

Comparison of bounds efficiency



Comparison of confidence envelopes



Local test and estimation of m_0

Let $\alpha \in [0, 1]$ and, for $A \subset \{1, \dots, m\}$, $H_A := \bigcap_{i \in A} H_i$ the intersection hypothesis for A .

Suppose to have at hand, for all $A \subset \{1, \dots, m\}$, a local test φ_A (a test for H_A), and denote $\mathcal{N}(X) = \{A \subset \mathbb{N}_m : \varphi_A(X) = 0\}$.
 $\mathcal{N}(X)$ is the set of potential candidate for $\mathcal{H}_0(\mu)$.

Proposition 3 (Inversion procedure, Genovese and Wasserman, 2006)

Suppose that $X \sim \mu$ and $\varphi_{\mathcal{H}_0(\mu)}$ is a test of level α . Then, the statistic $\hat{V}_{\mathcal{N}}$ defined by $\hat{V}_{\mathcal{N}}(S) := \max_{A \in \mathcal{N}} |A \cap S|$ is a confidence envelope of level α .

- **Advantage** : Best bound on $V(S)$ from local tests,
- **Drawback** : Exponential computational cost.

Inversion procedure and concentration inequalities

Suppose that for all $A \subset \{1, \dots, m\}$, $\varphi_A = \sup_{t \in [0,1]} \varphi_{A,t}$.

Example 3

Let $\alpha \in (0, 1]$ and $A \subset \llbracket 1, m \rrbracket$. From Bretagnolle's inequality,

$$\varphi_{A,t} = \mathbb{1} \left\{ \sqrt{|A|} (\hat{F}_A - \bar{F}_A)(t) > \lambda_\alpha \right\} = \mathbb{1} \left\{ \sum_{i \in A} \mathbb{G}_i(t) > \sqrt{|A|} \lambda_\alpha \right\},$$

where $\mathbb{G}(t) = (\mathbb{1}_{\{p_i \leq t\}} - F_i(t))_{1 \leq i \leq m}$, is a local test of level α .

Lemma 1

For all $S \subset \mathbb{N}_m$,

$$\hat{V}_{\mathcal{N}}(S) = \max \left\{ \ell \in \{0, \dots, |S|\} : \min_{A: |A \cap S| = \ell} \sup_{t \in [0,1]} \varphi_{A,t} = 0 \right\}.$$

Shortcuts and application to Bretagnolle's inequality

Lemma 2

Denote by $\varphi_{S,\ell}^{SC1} := \max_{t \in [0,1]} \min_{A: |A \cap S| = \ell} \varphi_{A,t}$, then :

$$\begin{aligned} \hat{V}_{\mathcal{N}}(S) &\leq \max \left\{ \ell \in \{0, \dots, |S|\} : \varphi_{S,\ell}^{SC1} = 0 \right\} \\ &\leq \min_{t \in [0,1]} \left\{ \max \left\{ \ell \in \{0, \dots, |S|\} : \min_{A: |A \cap S| = \ell} \varphi_{A,t} = 0 \right\} \right\}. \end{aligned}$$

Example 4

With Bretagnolle's local test, for $t \in [0, 1]$,

$$\min_{A: |A \cap S| = \ell} \varphi_{A,t} = \min_{0 \leq j \leq m - |S|} \mathbb{1} \left\{ \sum_{1 \leq i \leq \ell} (\mathbb{G}(t))_{(S:i)} + \sum_{1 \leq i \leq j} (\mathbb{G}(t))_{(S^c:i)} > \lambda_{\alpha} \sqrt{\ell + j} \right\}.$$

- Reduces computation cost to $\mathcal{O}(m^2 \log(m))$.

Upper bound on m_0 via Bretagnolle's inequality

Theorem 3 (Bretagnolle's upper bound on m_0)

Let $\alpha \in (0, 1]$. Suppose that for all $k \in \mathcal{K}$, $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda_\alpha = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$, and

$$b(t) = \max \left\{ \ell : \ell^{-1/2} \left(\sum_{1 \leq j \leq \ell} (\mathbb{G}(t))_{(j)} \right) \leq \lambda_\alpha \right\}, \text{ for all } t \in [0, 1].$$

Then, within the event $\{\mathcal{H}_0 \in \mathcal{N}\}$, we have $m_0 \leq \min_{t \in [0, 1]} b(t) := \hat{m}_0$.

Sketch of proof :

- Use Example 4 with $S = \{1, \dots, m\}$,
- Apply Lemma 2, second inequality.

Adaptive calibration

Theorem 4 (Adaptive version of Bretagnolle's top- k calibration)

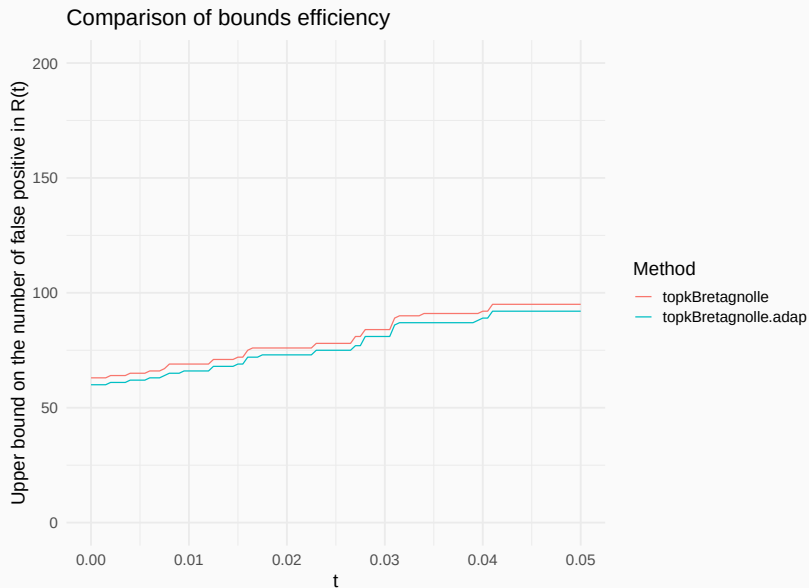
Let $\alpha \in (0, 1]$. Suppose that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p -values.

Define $\lambda_\alpha = \sqrt{\frac{1 + \log(1/\alpha)}{2}}$, $\mathbb{F}(t) := (F_i(t))_{1 \leq i \leq m}$, $\hat{m}_0$ as the estimator of m_0 from Theorem 3 and

$$\zeta_k = k \wedge \left\lfloor \sum_{1 \leq i \leq \hat{m}_0} \left(\mathbb{F}(p_{(k)}) \right)_{[i]} + \sqrt{\hat{m}_0} \lambda_\alpha \right\rfloor^2.$$

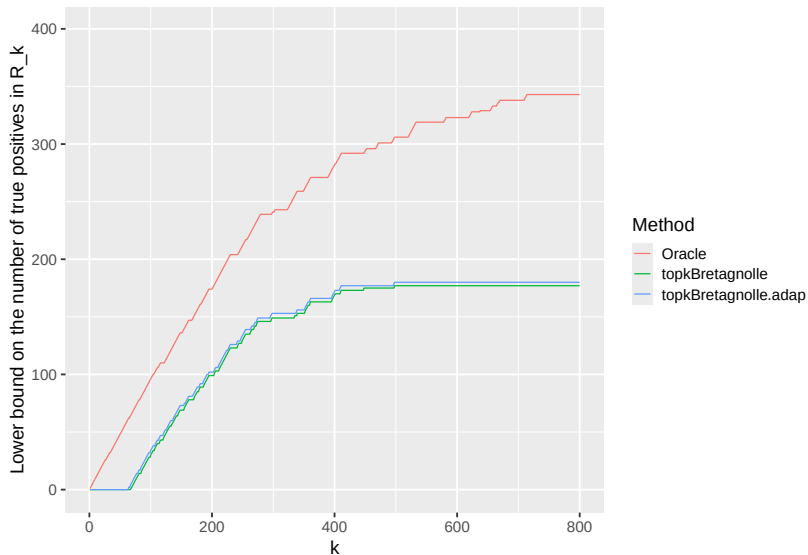
Then, the reference family $\mathfrak{R} = (R_k, \zeta_k)_{1 \leq k \leq m}$ has its JER bounded by α .

$^2x_{[i]}$ is the i -th largest coordinate of x .



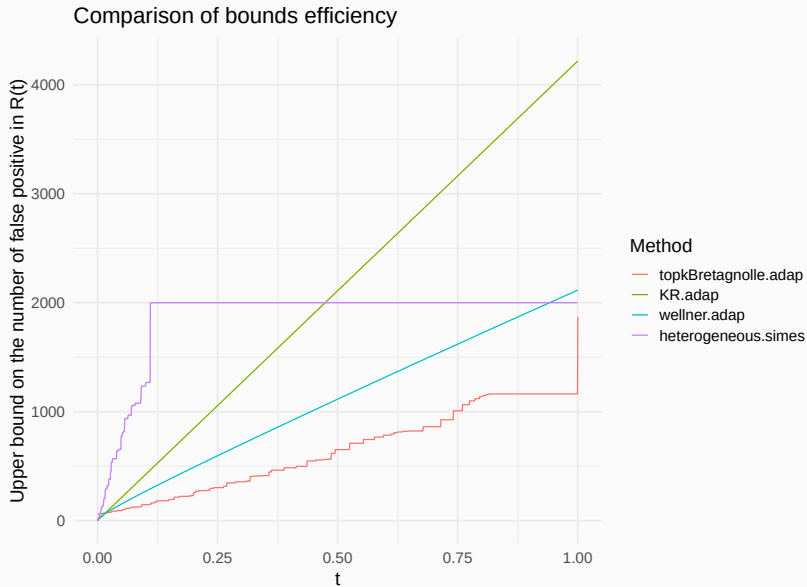
Simulations (2/2)

Comparison of confidence envelopes

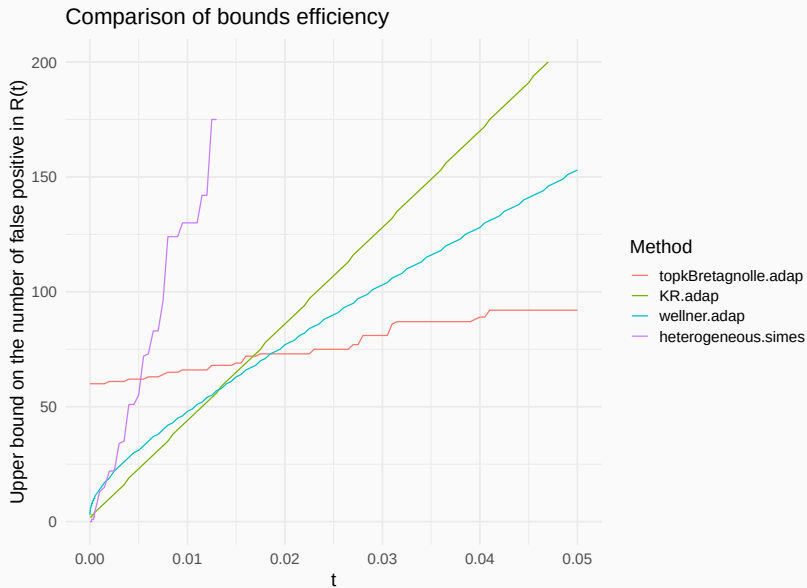


Comparison with existing literature

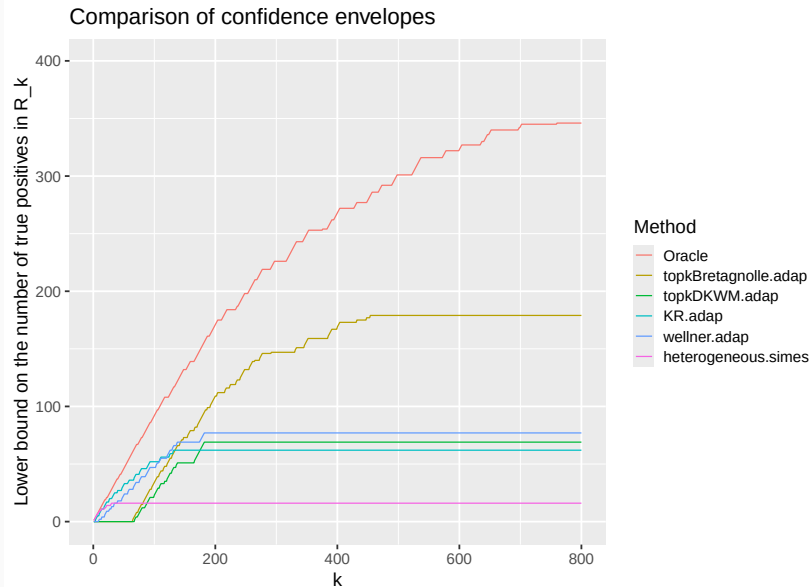
Bound efficiency comparisons (1/2)



Bound efficiency comparisons (2/2)

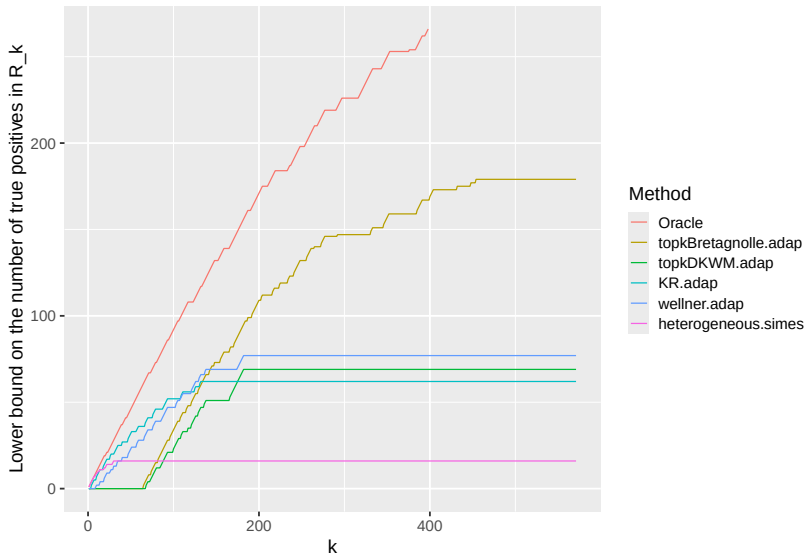


Envelope comparisons (1/2)



Envelopes comparisons (2/2)

Comparison of confidence envelopes



Summary :

- Applied a heterogeneous version of DKWM inequality in the context of post hoc MT,
- Get an inexact shortcut for the inversion procedure practically applicable to heterogeneous inequalities,
- Get an m_0 adaptive version of the resulting confidence envelope.

Perspectives :

- Numerical study of hybrid bounds,
- Get suitable heterogeneous versions of Wellner and K&R inequalities (work in progress).

Thanks a lot for your attention!

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Precisions on interpolated bounds

Interpolated confidence envelope

Proposition 4 (Interpolated bound, Blanchard et al., 2020)

Let $\mathfrak{R} = ((R_k, \zeta_k))_{1 \leq k \leq K}$ a family with JER bounded by $\alpha \in [0, 1]$ and

$$\mathcal{A}(\mathcal{R}) := \left\{ A \subset \mathbb{N}_m : \forall k \in \mathbb{N}_K, |R_k \cap A| \leq \zeta_k \right\}.$$

Then $V_{\mathfrak{R}}^*(S) := \max_{A \in \mathcal{A}(\mathcal{R})} |R \cap A|$ is a level α post hoc bound.

Proposition 5 (Interpolated bound nested setting)

Assume $(R_k)_k$ is a family of regions such that

$$\forall k \in \{1, \dots, K\}, R_k \subset R_{k+1}.$$

Then:

$$V_{\mathfrak{R}}^*(S) = \min_{1 \leq k \leq K} (\zeta_k + |S \setminus R_k|) \wedge |S|.$$

Complements on Uniform Wellner and K& R inequalities

Proposition 6

Let $\alpha \leq 0.31$. Then, in the top- k setting with p -values under the null independent,

$$\zeta_k := \left\lfloor \frac{\log(1/\alpha)}{\log(1 + \log(1/\alpha))} (1 + mp_{(k)}) \right\rfloor$$

gives a reference family with JER bounded by α .

Proposition 7 (Uniform version of Wellner inequality, Meah et al., 2024)

Let $U_1, \dots, U_m \stackrel{iid}{\sim} \mathcal{U}([0, 1])$ and $\kappa = \frac{\pi^2}{6}$. For all $\delta \in (0, 1]$,

$$\mathbb{P} \left(\forall t \in [0, 1], m^{-1} \sum_{1 \leq i \leq m} \mathbb{1}_{U_i \leq t} \leq th^{-1} \left(\frac{2 \log(\kappa/\delta) + 4 \log(1 + \log_2(1/t))}{mt} \right) \right) \geq 1 - \delta,$$

with $h(\lambda) = \lambda(\log \lambda - 1) + 1$.

Heterogeneous simultaneous k -FWER

Heterogeneous simultaneous k -FWER

Come back to homogeneous setting

Derive a simultaneous k -FWER inequality from a FDR control procedure

Example 5 (Simes inequality)

Let $\alpha \in [0, 1]$. Assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values. Then

$$\mathbb{P}\left(\exists i \in \{1, \dots, m_0\} : p_{(i:\mathcal{H}_0)} \leq \frac{\alpha i}{m}\right) \leq \alpha.$$

Derivation from BH procedure:

$$\begin{aligned}\mathbb{P}\left(\exists i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} \leq \frac{\alpha i}{m_0}\right) &= \mathbb{P}\left(\left|R^{\text{BH}}((p_i)_{i \in \mathcal{H}_0})\right| > 0\right) \\ &= \text{FWER}\left(R^{\text{BH}}((p_i)_{i \in \mathcal{H}_0})\right) \text{ since each } i \in \mathcal{H}_0 \text{ is true,} \\ &= \text{FDR}\left(R^{\text{BH}}((p_i)_{i \in \mathcal{H}_0})\right) \\ &\leq \alpha, \text{ by BH procedure.}\end{aligned}$$

Implies $JER(\mathfrak{R}) \leq \alpha$ for $R_k = \{i : p_i \leq \frac{\alpha k}{m}\}$ and $\zeta_k = k - 1$.

Definition 3

Let $\tau = (\tau_i)_{i \in \llbracket 1, m \rrbracket}$ be a family of thresholds.

Then the step up procedure associated to τ return

$$R^{\text{SU}(\tau)} = \{i \in \llbracket 1, m \rrbracket : p_i \leq \tau_{\hat{k}}\} \text{ and } \hat{k}^{\text{SU}(\tau)} = \max\{k : p_{(k)} \leq \tau_k\}.$$

The BH procedure is a **step-up** procedure.

Heterogeneous simultaneous k -FWER

Heterogeneous k -FWER method

A heterogeneous FDR control procedure

Theorem 5 (Döhler et al. (2018), Theorem 1)

Assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values. Furthermore, assume that the two families $(p_i)_{i \in \mathcal{H}_0}$ and $(p_i)_{i \in \mathcal{H}_1}$ are independent. Let τ be a threshold family such that $\tau_m < 1$. Then

$$\text{FDR} \left(R^{\text{SU}(\tau)} \right) \leq \max_{1 \leq k \leq m} \left\{ \max_{\substack{A \subset \llbracket 1, m \rrbracket \\ |A| = m - k + 1}} \left\{ \frac{1}{k} \sum_{i \in A} \frac{F_i(\tau_k)}{1 - F_i(\tau_m)} \right\} \right\}^{(*)}.$$

Let, for all $(s, t) \in [0, 1) \times [0, 1]$,

$$\mathbf{H}(s, t) = \left(\frac{F_i(t)}{1 - F_i(s)} \right)_{1 \leq i \leq m},$$

for $i \in \llbracket 1, m \rrbracket$, $\mathcal{A}_i$ be the support of p_i and

$$\mathcal{A} = \{0\} \bigcup_{i \in \llbracket 1, m \rrbracket} \mathcal{A}_i.$$

Threshold family construction

Theorem 6 (Heterogeneous Simes-like inequality)

Let τ be a threshold family satisfying $(\star) \leq \alpha$, and assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values. Then, the following inequality holds :

$$\mathbb{P}(\exists i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} \leq \tau_i) \leq \alpha.$$

Example 6 (Heterogeneous step-up procedure)

Let $\alpha \in (0, 1)$. Set

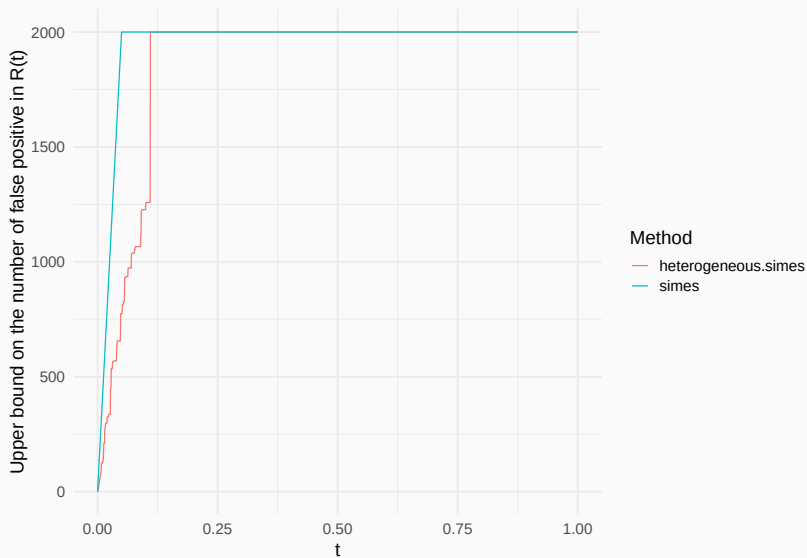
$$\tau_m \in \left\{ t \in \mathcal{A} : t < 1 \text{ and } (\mathbf{H}(t, t))_{[1]} \leq m\alpha \right\},$$

and, for all $k \in \llbracket 1, m-1 \rrbracket$, let

$$\tau_k = \max \left\{ t \in \mathcal{A}, t \leq \tau_m : (\mathbf{H}(\tau_m, t))_{[1]} + \cdots + (\mathbf{H}(\tau_m, t))_{[m-k+1]} \leq k\alpha \right\}.$$

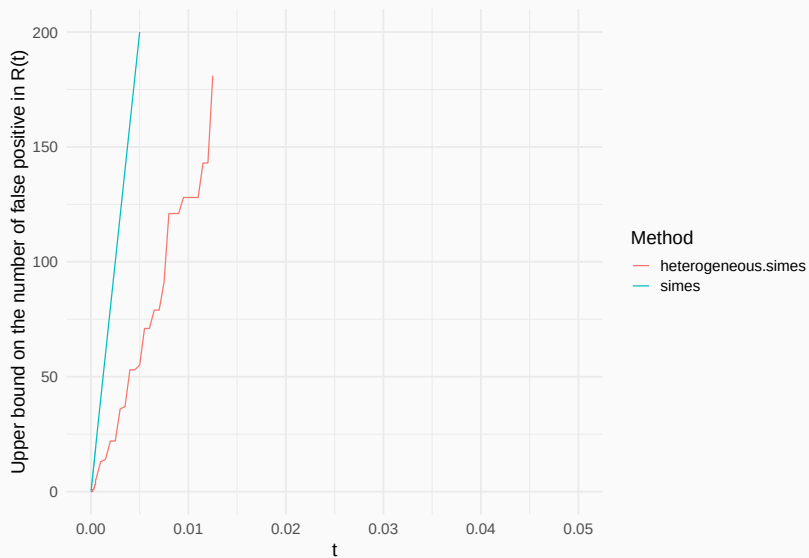
Simulations (1/2)

Comparison of bounds efficiency



Simulations (2/2)

Comparison of bounds efficiency



Heterogeneous simultaneous k -FWER

**Adaptive (in m_0) heterogeneous k -FWER
method**

Proposition 8

From the proof of Theorem 5 in Döhler et al. (2018) the following slightly more refined result can be derived:

$$\text{FDR} \left(R^{\text{SU}(\tau)} \right) \leq \max_{1 \leq k \leq m} \left\{ \max_{\substack{A \subset \llbracket 1, m \rrbracket \\ |A| = m - k + 1}} \left\{ \frac{1}{k} \sum_{i \in A \cap \mathcal{H}_0} \frac{F_i(\tau_k)}{1 - F_i(\tau_m)} \right\} \right\}^{(*)}.$$

Example 7

Let $\alpha \in (0, 1)$ and τ_m under the same constraints than in Example 6 and, for all $k \in \llbracket 1, m - 1 \rrbracket$, let

$$\tilde{\tau}_k = \max \left\{ t \in \mathcal{A}, t \leq \tilde{\tau}_m : \left(\mathbf{H}(\tilde{\tau}_m, t) \right)_{[1]} + \cdots + \left(\mathbf{H}(\tilde{\tau}_m, t) \right)_{[(m-k+1) \wedge m_0 : \mathcal{H}_0]} \leq k\alpha \right\}.$$

Then $\tilde{\tau}$ satisfies $(\star) \leq \alpha$.

Theorem 7 ($\hat{m}_0$ -adaptivity of a threshold family)

Let $\tilde{\tau}$ be a threshold family such that $(\star) \leq \alpha$. Furthermore, assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values. Then, for all (possibly data-dependant) threshold family τ such that $\tau_k \leq \tilde{\tau}_k$ for all $k \in \llbracket 1, m \rrbracket$ under the event $\left\{ \forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i \right\}$, the following inequality holds :

$$\mathbb{P}(\exists i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} \leq \tau_i) \leq \alpha.$$

Corollary 1 (Heterogeneous Simes like adaptive inequality)

Let $\tilde{\tau}$ be defined by Example 7 and $\hat{m}_0$ be a statistic such that

$$\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\} \subset \{m_0 \leq \hat{m}_0\}.$$

Furthermore, assume that $(p_i)_{i \in \mathcal{H}_0}$ is a family of independent p -values. Then the thresholds family defined by $\tau_m = \tilde{\tau}_m$ and $\forall k \in \llbracket 1, m-1 \rrbracket$,

$$\tau_k = \max \left\{ t \in \mathcal{A}, t \leq \tau_m : \left(\mathbf{H}(\tau_m, t) \right)_{[1]} + \cdots + \left(\mathbf{H}(\tau_m, t) \right)_{[(m-k+1) \wedge \hat{m}_0]} \leq k\alpha \right\}$$

verifies the inequality

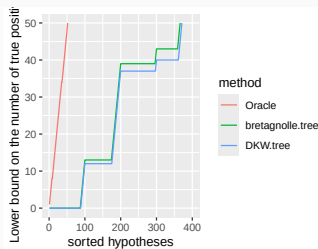
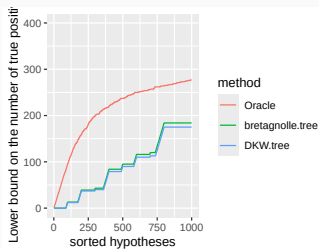
$$\mathbb{P}(\exists i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} \leq \tau_i) \leq \alpha.$$

Simulations settings

Same simulation settings than Döhler et al., 2018, with $N = 50$, $m = 2000$, $\pi_0 = 0.8$ and the response under H_1 simulated by binomials of parameter 0,25 against 0.1 under H_0 .

Deterministic setting simulations

Deterministic setting : the R_k are deterministic and have a forest structure, meaning that for all $(k, k') \in \{1, \dots, K\}^2$, $R_k \cap R_{k'} \in \{\emptyset, R_k, R_{k'}\}$.



Old simulations

- **top- k** : $m = 3200$ and $m_0 = 2400$. Continuous p -values sampled from a Gaussian model, with mean equal to 0 under H_0 and 3 under H_1 . Each p -value is discretized using a support of length 50 sampled from a uniform law.
- **k-FWER** : Same simulation setting with the support equal to 300.
- **Deterministic setting** : $m = 3200$ and $m_0 = 2800$, 8 active leaves of length 100 and each active leaf contains 50% of p -values under H_1 .

Comparison with bounds found in the literature

